DIGI Node — Tier 3: Populated PCB

A fully assembled PCB, soldered and bench-tested with its factory antennas. Meshtastic is flashed on the radio, and the radio's serial link to the node is already configured.

In the box

- Populated 82 mm PCB with an 8 GB microSD fitted
- Factory U.FL antennas: flat 2.4 GHz film, coiled LoRa whip, ceramic GPS patch

You supply: power over USB or your own regulated 5 V supply, and a housing if you want one.

Set it up

1. Fit the three U.FL antennas:
   - **GPS patch** → GPS module
   - **Long black whip** → LoRa radio
   - **Film antenna marked "2.4GHz"** → XIAO ESP32-S3

   Hold each connector parallel to the board, line up the rims and press straight down until it clicks. Pressing at an angle can crush the socket, and a crushed socket means replacing the module.
2. Flash AntiHunter on the ESP32 over USB with the web flasher at lukeswitz.github.io/AntiHunter. Choose **Full** for the web UI or **Headless** for quiet field operation.
3. Set the LoRa region on the radio. The region is **UNSET** when the unit leaves the bench. Until you set it the radio only receives, and the node will not answer over mesh.
4. Change the radio's Bluetooth pairing PIN, because every unit leaves the bench with the same one.
5. Make your own encrypted Meshtastic channel the primary channel and turn the public default channel off.
6. **Full firmware only:** join the `Antihunter` Wi-Fi network (password `antihunt3r123`), open `http://192.168.4.1` and change these credentials before doing anything else.
7. Set an erase key. Erase commands are refused until you do. Full: System tab → **Secure Data Destruction**. Headless: the **Erase key** field in the web flasher.

> **Never power the node without the LoRa antenna attached.**

Power

- Use a regulated 5 V supply only: USB, a bench supply set to 5 V, a proper DC-DC regulator or a UPS board.
- Never wire 18650 cells or an unregulated supply to the PCB.
- The node becomes unstable below 4.5 V.
- Connect **one USB cable at a time.** The ESP32 and the radio share the power rails.

Full guide

For each step above in detail, plus antennas, all detectors, the vibration sensor, mesh commands and Command Center, see the Operator's Guide: github.com/lukeswitz/AntiHunter/blob/main/docs/AntiHunter-Operators-Guide.pdf

- **Enclosure STLs, assembly manual:** github.com/lukeswitz/AntiHunter, `hw` folder
- **Firmware, docs, issues:** github.com/lukeswitz/AntiHunter
- **Command Center:** github.com/TheRealSirHaXalot/AntiHunter-Command-Control-PRO